

Tabela 17 – Features extracted with NFStream from dataset PCAP files.

N	Category	Feature	Description
1	Core Features	src_port	Transport-layer source port.
2		dst_port	Transport-layer destination port.
3		protocol	Transport-layer protocol.
4		bidirectional_first_seen_ms	Timestamp in milliseconds of the first bidirectional packet in the flow.
5		bidirectional_last_seen_ms	Timestamp in milliseconds of the last bidirectional packet in the flow.
6		bidirectional_duration_ms	Bidirectional flow duration in milliseconds.
7		bidirectional_packets	Bidirectional packet count.
8		bidirectional_bytes	Bidirectional byte count.
9		src2dst_first_seen_ms	Timestamp in milliseconds of the first src2dst packet.
10		src2dst_last_seen_ms	Timestamp in milliseconds of the last src2dst packet.
11		src2dst_duration_ms	src2dst flow duration in milliseconds.
12		src2dst_packets	src2dst packet count.
13		src2dst_bytes	src2dst byte count.
14		dst2src_first_seen_ms	Timestamp in milliseconds of the first dst2src packet.
15		dst2src_last_seen_ms	Timestamp in milliseconds of the last dst2src packet.
16		dst2src_duration_ms	dst2src flow duration in milliseconds.
17		dst2src_packets	dst2src packet count.
18		dst2src_bytes	dst2src byte count.
19	Post-Mortem Statistics	bidirectional_min_ps	Minimum bidirectional packet size.
20		bidirectional_mean_ps	Mean bidirectional packet size.
21		bidirectional_stddev_ps	Standard deviation of bidirectional packet size.
22		bidirectional_max_ps	Maximum bidirectional packet size.
23		src2dst_min_ps	Minimum src2dst packet size.
24		src2dst_mean_ps	Mean src2dst packet size.
25		src2dst_stddev_ps	Standard deviation of src2dst packet size.
26		src2dst_max_ps	Maximum src2dst packet size.
27		dst2src_min_ps	Minimum dst2src packet size.
28		dst2src_mean_ps	Mean dst2src packet size.
29		dst2src_stddev_ps	Standard deviation of dst2src packet size.
30		dst2src_max_ps	Maximum dst2src packet size.
31		bidirectional_min_piat_ms	Minimum bidirectional packet inter-arrival time.
32		bidirectional_mean_piat_ms	Mean bidirectional packet inter-arrival time.
33		bidirectional_stddev_piat_ms	Standard deviation of bidirectional packet inter-arrival time.
34		bidirectional_max_piat_ms	Maximum bidirectional packet inter-arrival time.
35		src2dst_min_piat_ms	Minimum src2dst packet inter-arrival time.
36		src2dst_mean_piat_ms	Mean src2dst packet inter-arrival time.
37		src2dst_stddev_piat_ms	Standard deviation of src2dst packet inter-arrival time.
38		src2dst_max_piat_ms	Maximum src2dst packet inter-arrival time.
39		dst2src_min_piat_ms	Minimum dst2src packet inter-arrival time.
40		dst2src_mean_piat_ms	Mean dst2src packet inter-arrival time.
41		dst2src_stddev_piat_ms	Standard deviation of dst2src packet inter-arrival time.
42		dst2src_max_piat_ms	Maximum dst2src packet inter-arrival time.
43		bidirectional_syn_packets	Number of bidirectional SYN packets.
44		bidirectional_cwr_packets	Number of bidirectional CWR packets.
45		bidirectional_ece_packets	Number of bidirectional ECE packets.
46		bidirectional_urg_packets	Number of bidirectional URG packets.
47		bidirectional_ack_packets	Number of bidirectional ACK packets.
48		bidirectional_psh_packets	Number of bidirectional PSH packets.
49		bidirectional_rst_packets	Number of bidirectional RST packets.
50		bidirectional_fin_packets	Number of bidirectional FIN packets.
51		src2dst_syn_packets	Number of src2dst SYN packets.
52		src2dst_cwr_packets	Number of src2dst CWR packets.
53		src2dst_ece_packets	Number of src2dst ECE packets.
54		src2dst_urg_packets	Number of src2dst URG packets.
55		src2dst_ack_packets	Number of src2dst ACK packets.
56		src2dst_psh_packets	Number of src2dst PSH packets.
57		src2dst_rst_packets	Number of src2dst RST packets.
58		src2dst_fin_packets	Number of src2dst FIN packets.
59		dst2src_syn_packets	Number of dst2src SYN packets.
60		dst2src_cwr_packets	Number of dst2src CWR packets.
61		dst2src_ece_packets	Number of dst2src ECE packets.
62		dst2src_urg_packets	Number of dst2src URG packets.
63		dst2src_ack_packets	Number of dst2src ACK packets.
64		dst2src_psh_packets	Number of dst2src PSH packets.
65		dst2src_rst_packets	Number of dst2src RST packets.
66		dst2src_fin_packets	Number of dst2src FIN packets.
67	Ground Truth	application_name	Application name identified by DPI.
68		application_category_name	Application category identified by DPI.
69		application_is_guessed	Indicates whether the classification was based on heuristics.
70		application_confidence	Underlying detection method used for classification.